DUNKL MATCHING-POLYNOMIAL DUALITY AND COMPLETE-GRAPH ALGEBRA PRESENTATION

DARIJ GRINBERG, ANATOL KIRILLOV, BORIS SHAPIRO

Editorial note. One principal foundation or monomial-congruence object. Complete original mathematical document89–2827, all native supporting proofs and manual bibliography retained. Raw source/archive/assets unchanged.

Editorial note. Original358–363 states bulk writing by GPT5.6 from old notes and tactical prompting, explicitly crediting all Section5 proofs to GPT5.6. Full literal disclosure is retained. Only foundational Sections3–4 targets are counted here; no AI-free claim or inference of their core origin is made. Planned Section5 saturation targets are not counted. Unselected Section5 statements remain author context with the explicit original proof credit.

Editorial note. Original field/orientation, coefficient-pairing normalization and characteristic-zero presentation assumptions retained exactly. Complete-graph duality/Specht/Hilbert/presentation forms are one object.

Editorial note. Authors retain responsibility for mathematics. No proof repair, new assumptions, independent proof review, symbolic/numerical replay, word/graph/counterexample search or certification.

Allowed standard prerequisites (editorial).

- Original graph matching/Dunkl/Specht framework or Weyl action/lattice-path/word framework. All native field, characteristic, orientation, pairing and word/move definitions retained verbatim; prior facts stay attributed in the complete original context.. Complete source-local selected proof material supplied without reconstructing external proofs, correcting hypotheses or independently certifying mathematics.

1. Introduction

Let $G = (V, E)$ be a finite simple graph. The matching generating polynomial of G records the numbers of sets of pairwise vertex-disjoint edges. In this note we lift this elementary invariant to a graded algebra

$$\mathcal{M}(G) = \mathbf{k}[u_e : e \in E]/(u_e u_f : e \cap f \neq \varnothing).$$

In this algebra, we define the signed incidence sums

$$\theta_v = \sum_{e=(v,w)} u_e - \sum_{e=(w,v)} u_e \qquad \text{for all } v \in V,$$

where an orientation of G has been chosen. We denote by $\mathcal{D}(G)$ the subalgebra generated by the θ_v and call it the *Dunkl matching algebra*. The goal of the present note is to understand this algebra $\mathcal{D}(G)$, in particular its Hilbert series.

Date: arxiv:2609.34260v2; CC0 1.0.

The homogeneous parts $\mathcal{D}(G)_k$ of $\mathcal{D}(G)$ stand in a natural duality to certain subspaces of the polynomial ring $\mathbf{k}[x_v \mid v \in V]$. Namely, for each matching M, define the polynomial

$$p_M = p_M(x) = \prod_{(i,j)\in M} (x_i - x_j) \in \mathbf{k}[x_v \mid v \in V].$$

For any $k \geq 0$, we define the vector subspace

$$P_k(G) := \operatorname{span}_{\mathbf{k}}\{p_M : M \text{ is a } k\text{-matching of } G\} \qquad \text{of } \mathbf{k}[x_v \mid v \in V].$$

The main results we shall show are the following:

- For each $k \geq 0$, we have
$$\dim \mathcal{D}(G)_k = \dim P_k(G);$$
see Theorem 3.2. More precisely, the proof gives a canonical perfect pairing between $\mathcal{D}(G)_k$ and $P_k(G)$ (see Remark 3.3); thus, $P_k(G)$ becomes a dual polynomial model for $\mathcal{D}(G)_k$. Consequently, the Hilbert function of $\mathcal{D}(G)$ records the ranks of the matching-restricted products $\prod_{(i,j)\in M}(x_i - x_j)$.
- For the complete graph K_n, the dual polynomial model becomes classical representation theory: for $0 \leq k \leq \lfloor n/2 \rfloor$,
$$P_k(K_n) \cong S^{(n-k,k)}.$$
Here $S^{(n-k,k)}$ is the Specht module of shape $(n-k,k)$ (and is irreducible in characteristic zero). Thus, the Hilbert series of $\mathcal{D}(K_n)$ is the generating function of the n-th row of the Catalan triangle; see Theorem 4.1. In characteristic zero, we also obtain the presentation
$$\mathcal{D}(K_n) \cong \mathbf{k}[z_1, \ldots, z_n]/(z_1 + \cdots + z_n, z_1^2, \ldots, z_n^2);$$
see Theorem 4.4.
- Motivated by the complete-graph case, we say that a graph G is *k-saturated* when $P_k(G) = P_k(K_n)$. We prove the implications
$$k\text{-linked} \implies k\text{-saturated} \implies k\text{-connected};$$
see Propositions 5.4 and 5.11. In degree 2 the second implication is an equivalence:
$$P_2(G) = P_2(K_n) \iff G \text{ is 2-connected}$$
for $n \geq 4$; see Theorem 5.9. We also show that if $\overline{G}$ is a matching, then G is saturated in every possible degree; see Theorem 5.12.
- The matching generating polynomial and the Dunkl Hilbert series of G (that is, the Hilbert series of $\mathcal{D}(G)$) are incomparable invariants: neither determines the other. Indeed, P_3 and K_3 have the same Dunkl Hilbert series $1 + 2t$ but different matching generating polynomials $1 + 2t$ and $1 + 3t$, respectively; in the other direction, Proposition 5.17 gives graphs with the same matching generating polynomial but different Dunkl Hilbert series. We also obtain explicit positive formulas in sparse cases. Namely,
$$\mathcal{D}(G) = \mathcal{M}(G) \iff G \text{ is a forest};$$

see Proposition 5.15. For a unicyclic graph with unique cycle C and $F = G - V(C)$, we obtain a short exact sequence of graded vector spaces

$$0 \longrightarrow \mathcal{M}(F)(-1) \longrightarrow \mathcal{M}(G) \longrightarrow \bigoplus_{k\geq 0} P_k(G) \longrightarrow 0,$$

and hence (using the notation $\mathfrak{m}_G$ for the matching generating polynomial of G)

$$\mathrm{Hilb}(\mathcal{D}(G), t) = \mathfrak{m}_G(t) - t\mathfrak{m}_F(t);$$

see Theorem 5.16. Minimal pairs with the same matching generating polynomial but different Dunkl Hilbert series occur on 4 vertices in general, on 5 vertices among connected graphs, and on 6 vertices among 2-connected graphs; see Proposition 5.17 and Remark 5.18.

- Finally, we survey a related construction in the larger square-zero edge algebra [24], where distinct incident edge generators are allowed to multiply. For each nontrivial vertex cut $\delta(S)$, consider the cut monomial

$$\eta_S := \prod_{e\in\delta(S)} u_e,$$

and impose the relations $\eta_S = 0$ for all such S. If these relations are imposed inside the subalgebra generated by the signed incidence sums, the resulting quotient is the central graphical zonotopal algebra. If they are imposed instead in the whole square-zero edge algebra, the surviving monomials are precisely the u_F for which deleting F leaves G connected; hence the Hilbert series counts connected spanning subgraphs and is a specialization of the all-terminal reliability polynomial. See Propositions A.1 and A.2; these results are not new, so we provide no proofs.

Motivation. This work is inspired by the Fomin–Kirillov approach to Schubert calculus, see [10, 2, 18]. However in our set-up the adjective "Dunkl" is motivational only. Unlike the Fomin–Kirillov algebra [10], our matching algebra is commutative and kills every product of two incident edge generators.

Recall that the Fomin–Kirillov quadratic algebra $\mathcal{E}_n$ is generated by antisymmetric symbols $[ij] = -[ji]$ subject, in particular, to $[ij]^2 = 0$, commutation of generators with disjoint supports, and certain three-term relations on triples of indices. Its distinguished elements

$$\vartheta_i = \sum_{j\neq i} [ij]$$

are known as the Dunkl elements. A fundamental feature of the construction is that these elements commute, and the commutative subalgebra which they generate provides an algebraic model for the cohomology ring of the complete flag variety; see [10]. This point of view has subsequently led to variants for other Coxeter groups, quantum cohomology, K-theory, and Nichols–Woronowicz algebras.

Blasiak, Liu, and Mészáros [7, Definition 2.3] have introduced restrictions of this algebra to a graph. For a graph G on $[n]$, they define $\mathcal{E}_G$ to be the subalgebra of $\mathcal{E}_n$ generated by those $[ij]$ for which $\{i, j\}$ is an edge of G. Their graph restriction

is thus a subalgebra construction, rather than the quotient obtained by setting nonedge generators equal to zero.

Our construction may be viewed as a quotient of this picture. Give the generator corresponding to an unoriented edge $\{i,j\}$ the sign determined by an orientation and write it as $u_{ij} = -u_{ji}$. If, in addition to the Fomin–Kirillov relations, one kills every product

$$u_{ij}u_{k\ell} \qquad \text{with} \qquad \{i,j\} \cap \{k,\ell\} \neq \varnothing,$$

and kills the generators corresponding to nonedges of G, then the resulting commutative quotient is precisely the matching algebra $\mathcal{M}(G)$. Under this quotient the Dunkl element ϑ_i descends to the signed incidence sum θ_i. Thus $\mathcal{D}(G)$ can be interpreted as the part of the Dunkl-element construction which survives after only products supported on matchings are retained. Equivalently, the quotient map $\mathcal{E}_n \twoheadrightarrow \mathcal{M}(K_n)$ that kills products of incident edge generators restricts to a surjection $\mathcal{E}_G \twoheadrightarrow \mathcal{M}(G)$, where $\mathcal{M}(G)$ is identified with the subalgebra of $\mathcal{M}(K_n)$ generated by the edges of G.

This truncation changes the nature of the problem considerably. The flag-variety and Bruhat-order connections of the classical setting are replaced by the incidence structure of a graph, while the linear forms $x_i - x_j$, i.e., the roots of type A, reappear on the dual side. One may therefore ask how much of the structure carried by Dunkl elements survives under this severe matching truncation. The results below show that the answer retains a surprising amount of structure: for the complete graph the homogeneous components are governed by two-row Specht modules, whereas for arbitrary graphs the failure or success of spanning the corresponding Specht module detects connectivity properties of the graph. In this sense, the matching quotient provides a bridge between the Dunkl-element philosophy, matching theory, and the representation theory of the symmetric group.

Related work. Subalgebras generated by vertices in the commutative algebra generated by edges have been considered in e.g., [24, 25]. Moreover, in the case of complete graphs they also have a geometric interpretation, see [26]. Products of arbitrary subfamilies of a vector configuration have a rich theory related to Tutte polynomials; see Berget [5] and the zonotopal literature [1, 15]. Our restriction to vertex-disjoint subfamilies is not matroidal: it remembers the incidence of edges at vertices. It also differs from graph Specht modules and their relationship with matching polytopes studied by Liu [21]. The construction lies naturally between these two subjects.

Several nearby constructions attach graded objects to matchings but use a different algebraic mechanism. Numata [23] studies an Artinian Gorenstein algebra obtained by apolarity from a weighted matching generating function and proves a strong Lefschetz property. Li [19] proves strong equivariant log-concavity for the graded permutation representation on the matchings of an arbitrary graph. For perfect matchings of complete graphs, Filmus and Lindzey [11] develop harmonic polynomial models involving Specht polynomials and matching-inclusion matrices. Bedratyuk's graph algebra [4], whose basis is indexed by all simple graphs on a fixed vertex set, also carries Boolean up/down operators and two-row Specht modules. Bergeron, Chan, Soltani, and Zabrocki [3] study a construction deceptively parallel to our $P_k(K_n)$ in an exterior algebra: products of differences

$\xi_j - \xi_i$ indexed by noncrossing pairings give bases of exterior quasisymmetric harmonic spaces whose degree-k dimensions agree with $\dim P_k(K_n)$. We compare the two constructions more closely in Remark 4.3. None of these constructions is the incidence-generated subalgebra $\mathcal{D}(G) \subseteq \mathcal{M}(G)$ studied here.

There is one direct overlap at the level of the abstract complete-graph presentation. Jonsson Kling, Lundqvist, Mohammadi, Orth, and Sáenz-de-Cabezón [17] study the more general almost complete intersections

$$\mathbf{k}[z_1, \dots, z_n]/(z_1^2, \dots, z_n^2, (z_1 + \cdots + z_n)^q).$$

The quotient occurring in our characteristic-zero presentation of $\mathcal{D}(K_n)$ is their case $q = 1$. The contribution here is therefore not the abstract quotient itself, but its realization inside a matching algebra, its matching-product duality, and the resulting saturation problem for arbitrary graphs.

AI use statement. The bulk of this paper was written by GPT-5.6 based on old notes (which contained more questions than proofs) and some tactical prompting; in particular, GPT-5.6 contributed all the proofs in Section 5. We have read and verified the entire text and edited it for clarity and style.

Acknowledgments. We thank Per Alexandersson for inspiring conversations early on in the project.

Notations. Throughout, $\mathbf{k}$ is a field. In some places, we will additionally assume that $\mathbf{k}$ has characteristic zero.

2. The matching algebra

Let $G = (V, E)$ be a finite simple graph. Its edges are 2-element sets of vertices; thus, two edges e, f satisfy $e \cap f \neq \varnothing$ if and only if they have a common endpoint (this allows $e = f$). Two edges e and f are said to be *incident* if $e \cap f \neq \varnothing$.

Definition 2.1. The *matching algebra* of G is the commutative $\mathbf{k}$-algebra

$$\mathcal{M}(G) := \mathbf{k}[u_e : e \in E]/\big(u_e u_f : e \cap f \neq \varnothing\big).$$

That is, $\mathcal{M}(G)$ is the quotient of the polynomial ring $\mathbf{k}[u_e : e \in E]$ by the monomial ideal generated by all products $u_e u_f$ with e and f being two incident edges (including the case $e = f$). It is graded by degree, where every generator u_e has degree 1.

For a matching $M \subseteq E$ of G, define the element

$$u_M := \prod_{e \in M} u_e \in \mathcal{M}(G), \qquad \text{in particular } u_\varnothing := 1.$$

Proposition 2.2. *The elements u_M, indexed by the matchings M of G, form a homogeneous $\mathbf{k}$-basis of $\mathcal{M}(G)$. Consequently, the Hilbert series of $\mathcal{M}(G)$ is*

$$\operatorname{Hilb}(\mathcal{M}(G), t) = \sum_{k \geq 0} m_k(G) t^k,$$

where $m_k(G)$ is the number of k-matchings of G.

Proof. The defining ideal $\left(u_e u_f : e \cap f \neq \varnothing\right)$ is monomial. Its standard monomials are the square-free products $u_{e_1} u_{e_2} \cdots u_{e_k}$ of generators such that no two of the edges $e_1, e_2, \ldots, e_k$ have a common endpoint; these are exactly the u_M corresponding to the matchings M of G. Thus, these u_M (or, rather, their residue classes) form a basis of the quotient algebra $\mathcal{M}(G)$. The formula for the Hilbert series follows directly from this. $\square$

Thus, for the complete graph K_n, we have

$$\operatorname{Hilb}(\mathcal{M}(K_n), t) = \sum_{0 \leq k \leq n/2} \frac{n!}{2^k k! (n-2k)!} t^k,$$

and the complete bipartite graph $K_{a,b}$ satisfies

$$\operatorname{Hilb}(\mathcal{M}(K_{a,b}), t) = \sum_{k \geq 0} k! \binom{a}{k} \binom{b}{k} t^k.$$

In particular, $\dim \mathcal{M}(K_n)$ is the number of involutions in S_n.

Equivalently, if $\Delta(G)$ denotes the matching complex of G (the independence complex of the line graph $L(G)$), then $\mathcal{M}(G)$ is the Artinian monomial quotient obtained from its Stanley–Reisner ring $\mathbf{k}[\Delta(G)]$ by adjoining the squares u_e^2 to the Stanley–Reisner ideal. Thus $\mathcal{M}(G)$ itself is not a Stanley–Reisner ring in the usual sense. Matching complexes and their face rings are classical; see, for example, Björner–Lovász–Vrećica–Živaljević [6] and Jonsson [16, Chapter 5]. Ferrarello and Fröberg [9] explicitly relate the Hilbert series of the ordinary Stanley–Reisner ring of a matching complex to graph-subgraph polynomials.

More directly, Dao and Nair [8] associate to an arbitrary simplicial complex Δ the square-zero face algebra

$$A(\Delta) := \mathbf{k}[x_1, \ldots, x_m] / (I_\Delta, x_1^2, \ldots, x_m^2),$$

and study its Lefschetz properties. Thus our matching algebra is exactly $\mathcal{M}(G) = A(\Delta(G))$. Such algebras are also a special case of the quadratic monomial quotients studied by Migliore, Nagel, and Schenck [22]. What we shall study (starting in the next section) is a certain **subalgebra** $\mathcal{D}(G)$ of $\mathcal{M}(G)$, which – to our knowledge – is new.

Example 2.3. Let $G = P_4$ be the path $1 - 2 - 3 - 4$. Then

$$\mathcal{M}(P_4) = \mathbf{k}[u_{12}, u_{23}, u_{34}] / (u_{12}^2, u_{23}^2, u_{34}^2, u_{12}u_{23}, u_{23}u_{34}).$$

Thus

$$1, \quad u_{12}, \quad u_{23}, \quad u_{34}, \quad u_{12}u_{34}$$

is a basis of $\mathcal{M}(P_4)$. The only nonzero product of two edge generators is $u_{12}u_{34}$, so

$$\operatorname{Hilb}(\mathcal{M}(P_4), t) = 1 + 3t + t^2, \qquad (u_{12} + u_{34})^2 = 2u_{12}u_{34}.$$

By contrast, for the triangle K_3 every two edges meet, so $\mathcal{M}(K_3)$ has basis $1, u_{12}, u_{13}, u_{23}$ and its positive-degree ideal has square zero.

3. Signed incidence sums and the Dunkl matching algebra

Fix an orientation of G. Define *incidence coefficients* $\varepsilon_{t,e}$ for all $e \in E$ and $t \in V$ as follows: If an edge e is oriented from v to w, let

$$\varepsilon_{v,e} = 1, \qquad \varepsilon_{w,e} = -1,$$

and let all other incidence coefficients $\varepsilon_{t,e}$ (with $t \in V \setminus \{v, w\}$) be zero.

Definition 3.1. For $v \in V$, define the *signed incidence sum*

$$\theta_v = \sum_{e \in E} \varepsilon_{v,e} u_e \in \mathcal{M}(G).$$

The *Dunkl matching algebra* is the graded subalgebra $\mathcal{D}(G)$ of $\mathcal{M}(G)$ generated by these signed incidence sums. In other words:

$$\mathcal{D}(G) := \mathbf{k}[\theta_v : v \in V] \subseteq \mathcal{M}(G).$$

Reversing the orientation of an edge amounts to replacing the corresponding generator u_e by $-u_e$; hence the isomorphism type of $\mathcal{D}(G)$ is orientation-independent. We have

$$\theta_v^2 = 0, \qquad \sum_{v \in C} \theta_v = 0 \tag{1}$$

for every vertex v and every connected component C of G.

Consider the polynomial ring $\mathbf{k}[x_v \mid v \in V]$. If an edge e is oriented as $e = (i, j)$, define the linear polynomial $p_e := x_i - x_j$. For a matching M of G, define the polynomial

$$p_M := \prod_{e \in M} p_e = \prod_{e=(i,j) \in M} (x_i - x_j), \qquad \text{so in particular } p_\varnothing := 1. \tag{2}$$

Reversing the orientation of an edge changes the corresponding p_e and p_M only by a sign, so all spans considered below are orientation-independent.

For each integer $k \geq 0$, we let $\mathrm{Match}_k(G)$ denote the set of all k-matchings of G (that is, of all matchings with exactly k edges), and we define the $\mathbf{k}$-vector subspace

$$P_k(G) := \mathrm{span}\{p_M : M \in \mathrm{Match}_k(G)\} \qquad \text{of } \mathbf{k}[x_v \mid v \in V]. \tag{3}$$

Consider the $\mathbf{k}$-vector space $\mathbf{k}^V$ with basis V; its standard basis vectors will be denoted by e_v. Let $S^k(\mathbf{k}^V)$ denote its kth symmetric power. Multiplication gives a linear map

$$\mu_k : S^k(\mathbf{k}^V) \longrightarrow \mathcal{M}(G)_k, \qquad e_{v_1} \cdots e_{v_k} \longmapsto \theta_{v_1} \cdots \theta_{v_k},$$

whose image is precisely the k-th graded component $\mathcal{D}(G)_k$ of $\mathcal{D}(G)$.

Theorem 3.2 (Dunkl-polynomial duality)**.** *For every $k \geq 0$, we have*

$$\dim \mathcal{D}(G)_k = \dim P_k(G). \tag{4}$$

Moreover, for $\alpha = (\alpha_v)_{v \in V} \in \mathbb{N}^V$ with $|\alpha| = k$, write

$$e^\alpha := \prod_{v \in V} e_v^{\alpha_v} \in S^k(\mathbf{k}^V), \qquad x^\alpha := \prod_{v \in V} x_v^{\alpha_v} \in \mathbf{k}[x_v \mid v \in V].$$

The elements e^α form a basis of $S^k(\mathbf{k}^V)$. Let $(e^\alpha)^*$ denote the dual basis, and define the linear isomorphism

$$\iota_k : S^k(\mathbf{k}^V)^* \longrightarrow \mathbf{k}[x_v : v \in V]_k, \qquad (e^\alpha)^* \longmapsto x^\alpha.$$

Then, for every $M \in \mathrm{Match}_k(G)$, we have

$$\iota_k\left(\mu_k^*(u_M^*)\right) = p_M, \tag{5}$$

where $u_M^* \in (\mathcal{M}(G)_k)^*$ denotes extraction of the u_M-coefficient (i.e., the $\mathbf{k}$-linear map that sends each polynomial in $\mathcal{M}(G)_k$ to its u_M-coefficient).

Proof. Fix a k-matching M. We begin by proving (5). Observe that the map ι_k sends each form $f \in S^k(\mathbf{k}^V)^*$ to the polynomial $\sum_{|\alpha|=k} f(e^\alpha)x^\alpha$. Hence, in order to prove (5), it suffices to show that

$$\left(\mu_k^*(u_M^*)\right)(e^\alpha) = [x^\alpha]p_M \qquad \text{for every } \alpha \in \mathbb{N}^V \text{ satisfying } |\alpha| = k. \tag{6}$$

So let us fix a family $\alpha = (\alpha_v)_{v\in V} \in \mathbb{N}^V$ with $|\alpha| = k$. Then,

$$\left(\mu_k^*(u_M^*)\right)(e^\alpha) = u_M^*\left(\mu_k(e^\alpha)\right) = u_M^*\left(\prod_{v\in V}\theta_v^{\alpha_v}\right) = [u_M]\prod_{v\in V}\theta_v^{\alpha_v}$$

(where $[u_M]r$ denotes the coefficient of u_M in some $r \in \mathcal{M}(G)$). In order to prove (6), we must show that this scalar is exactly the coefficient $[x^\alpha]p_M$.

First suppose that $\alpha_v \geq 2$ for some vertex v. Then $\prod_{v\in V}\theta_v^{\alpha_v} = 0$, since $\theta_v^2 = 0$ by (1). Hence the left-hand side of (6) is zero. On the other hand, p_M is squarefree in the variables x_v, since no vertex belongs to two edges of M. Thus $[x^\alpha]p_M = 0$ as well. So the equality (6) holds in this case because both of its sides vanish.

It remains to consider the case when $\alpha_v \leq 1$ for each vertex v. Let

$$S := \{v \in V : \alpha_v = 1\},$$

so that $|S| = |\alpha| = k$ and

$$\prod_{v\in V}\theta_v^{\alpha_v} = \prod_{v\in S}\theta_v = \prod_{v\in S}\left(\sum_{e\in E}\varepsilon_{v,e}u_e\right). \tag{7}$$

The coefficient of u_M in this product is zero unless every edge of M has an endpoint in S. This condition is equivalent to saying that every edge of M has **exactly one** endpoint in S (because the k edges of M are pairwise disjoint, and thus choosing one endpoint of each of them already gives $k = |S|$ distinct vertices, leaving no space for second endpoints).

Assume this condition holds. Then there is a unique way to obtain u_M when expanding the product (7): for each $e \in M$, choose u_e from the factor indexed by the unique vertex $v \in e \cap S$. The coefficient of u_M in (7) is therefore

$$\prod_{\substack{e\in M;\\ v \text{ is the unique element of } e\cap S}} \varepsilon_{v,e}. \tag{8}$$

But the coefficient of x_v in p_e is exactly $\varepsilon_{v,e}$. Consequently the same product (8) is the coefficient of $x^\alpha = \prod_{v\in S} x_v$ in

$$p_M = \prod_{e\in M} p_e.$$

Thus (6) holds in the $\alpha_v \leq 1$ case, at least under the condition that every edge of M has an endpoint in S. But (6) also holds if this condition is violated, since both sides vanish in this case. Hence (6) always holds.

Forget that we fixed α. We thus have proved (6), which in turn completes the proof of (5).

Now forget that we fixed M. The forms u_M^* form a basis of $\mathcal{M}(G)_k^*$, whereas the p_M span $P_k(G)$. Hence, the equality (5) entails

$$\iota_k(\operatorname{im} \mu_k^*) = P_k(G),$$

so that $P_k(G) = \iota_k(\operatorname{im} \mu_k^*) \cong \operatorname{im} \mu_k^*$ (since ι_k is an isomorphism). Therefore

$$\dim P_k(G) = \dim \operatorname{im} \mu_k^* = \operatorname{rank} \mu_k^* = \operatorname{rank} \mu_k = \dim \operatorname{im} \mu_k = \dim \mathcal{D}(G)_k,$$

as claimed. □

Remark 3.3. The proof gives a perfect pairing

$$\mathcal{D}(G)_k \times P_k(G) \longrightarrow \mathbf{k}.$$

Indeed, for any linear map $T : X \to Y$ between two finite-dimensional $\mathbf{k}$-vector spaces X and Y, the natural pairing

$$\operatorname{im} T \times \operatorname{im} T^* \to \mathbf{k}, \qquad (Tx, T^*\varphi) \mapsto \varphi(Tx) = (T^*\varphi)(x)$$

between $\operatorname{im} T$ and $\operatorname{im} T^*$ is perfect, since

$$\operatorname{im} T \cong X/\ker T, \qquad \operatorname{im} T^* = (\ker T)^\perp \cong (X/\ker T)^*.$$

In the present case (where $X = S^k(\mathbf{k}^V)$ and $Y = \mathcal{M}(G)_k$ and $T = \mu_k$), we thus obtain a pairing $\langle \cdot, \cdot \rangle : \mathcal{D}(G)_k \times P_k(G) \to \mathbf{k}$ given explicitly by

$$\langle d, p_M \rangle = [u_M]d \qquad \text{for all } d \in \mathcal{D}(G)_k \text{ and } M \in \operatorname{Match}_k(G) \tag{9}$$

(by (5)). Thus, the spaces $\mathcal{D}(G)_k$ and $P_k(G)$ are in perfect duality, and in particular $\dim \mathcal{D}(G)_k = \dim P_k(G)$, as claimed.

The equality (4) says that $\operatorname{Hilb}(\mathcal{D}(G), t) = \sum_k \dim P_k(G) t^k$. Although $m_k(G)$ counts the chosen generators of $P_k(G)$, their linear relations contain additional information about G.

Example 3.4 (Small Dunkl matching algebras)**.** Orient every displayed edge from the smaller endpoint to the larger one (i.e., orient it as (i, j) where $i < j$).

For the path graph P_4, we have

$$\theta_1 = u_{12}, \qquad \theta_2 = -u_{12} + u_{23}, \qquad \theta_3 = -u_{23} + u_{34}, \qquad \theta_4 = -u_{34}.$$

Hence

$$u_{12} = \theta_1, \qquad u_{23} = \theta_1 + \theta_2, \qquad u_{34} = -\theta_4,$$

so every edge generator u_e already lies in $\mathcal{D}(P_4)$. Therefore

$$\mathcal{D}(P_4) = \mathcal{M}(P_4), \qquad \operatorname{Hilb}(\mathcal{D}(P_4), t) = 1 + 3t + t^2.$$

(The same argument shows $\mathcal{D}(G) = \mathcal{M}(G)$ for every forest G, since the incidence matrix of a forest has rank $|E|$.)

For the triangle K_3, we have

$$\theta_1 = u_{12} + u_{13}, \qquad \theta_2 = -u_{12} + u_{23}, \qquad \theta_3 = -u_{13} - u_{23}.$$

Thus $\theta_1 + \theta_2 + \theta_3 = 0$, and all products of two θ_i vanish because every pair of edges of K_3 is incident. Consequently

$$\mathcal{D}(K_3) = \mathbf{k} \oplus \operatorname{span}\{\theta_1, \theta_2\}, \qquad \operatorname{Hilb}(\mathcal{D}(K_3), t) = 1 + 2t,$$

so here $\mathcal{D}(K_3)$ is a proper subalgebra of $\mathcal{M}(K_3)$.

The first nontrivial degree-2 example is K_4. Put

$$m_1 = u_{12}u_{34}, \qquad m_2 = u_{13}u_{24}, \qquad m_3 = u_{14}u_{23}.$$

These are the three degree-2 basis elements of $\mathcal{M}(K_4)$. Direct multiplication gives

$$\theta_1\theta_2 = m_2 + m_3, \qquad \theta_1\theta_3 = m_1 - m_3, \qquad \theta_2\theta_3 = -m_1 - m_2.$$

The three displayed vectors have one linear relation and span a 2-dimensional space; since $\theta_4 = -(\theta_1 + \theta_2 + \theta_3)$ and $\theta_i^2 = 0$, they span all of $\mathcal{D}(K_4)_2$. Hence

$$\operatorname{Hilb}(\mathcal{D}(K_4), t) = 1 + 3t + 2t^2.$$

This is the smallest example in which the matching algebra has more degree-2 basis elements than the Dunkl matching algebra has degree-2 dimensions.

Proposition 3.5. *The following properties hold.*

(1) *If $G = G_1 \sqcup G_2$, then $\mathcal{D}(G) \cong \mathcal{D}(G_1) \otimes \mathcal{D}(G_2)$.*
(2) *We have $\mathcal{D}(G)_k = 0$ for all $k > \nu(G)$, where $\nu(G)$ is the matching number of G (that is, the size of a largest matching of G).*
(3) *We have $\dim \mathcal{D}(G)_1 = |V| - c(G)$, where $c(G)$ is the number of connected components of G.*
(4) *If H is a spanning subgraph of G, then*

$$P_k(H) \subseteq P_k(G), \qquad \dim \mathcal{D}(H)_k \leq \dim \mathcal{D}(G)_k.$$

Proof. The first assertion follows from the disjoint sets of variables and incidence sums. The second follows from (4). For the third, observe that $P_1(G)$ is the span of the vectors $x_i - x_j$, which are the columns of the incidence matrix of G; thus, its dimension is the rank of the incidence matrix, which is known to equal $|V| - c(G)$. The fourth assertion is immediate (note that there is a graded surjective $\mathbf{k}$-algebra homomorphism $\mathcal{D}(G) \to \mathcal{D}(H)$ given by sending all u_e with $e \notin E(H)$ to 0). □

Remark 3.6 (Dependence on the characteristic). Although Theorem 3.2 holds over every field, the Hilbert series of $\mathcal{D}(G)$ can depend on the characteristic. For example, let $G = C_5 \,\square\, K_2$ be the pentagonal prism, with vertices $a_0, \ldots, a_4, b_0, \ldots, b_4$ and edges

$$a_ia_{i+1}, \qquad b_ib_{i+1}, \qquad a_ib_i \qquad (i \in \mathbb{Z}/5\mathbb{Z}).$$

Then

$$\operatorname{Hilb}(\mathcal{D}(G), t) = 1 + 9t + 35t^2 + 75t^3 + \alpha_{\mathbf{k}}t^4 + 11t^5, \tag{10}$$

where $\alpha_{\mathbf{k}} = 79$ if $\operatorname{char} \mathbf{k} = 2$, and $\alpha_{\mathbf{k}} = 80$ otherwise. This shows that the inclusion $\mathcal{D}(G) \to \mathcal{M}(G)$ need not remain injective after reduction modulo a prime.

4. Complete graphs and two-row Specht modules

We now specialize to the complete graph K_n on the vertex set $[n]$. The symmetric group S_n acts on $\mathbf{k}[x_1, \dots, x_n]$ by permuting the variables:

$$\sigma(x_i) = x_{\sigma(i)} \qquad (\sigma \in S_n).$$

Since S_n also permutes the k-matchings of K_n, this action preserves $P_k(K_n)$: more precisely, $\sigma(p_M) = \pm p_{\sigma(M)}$, where the sign only reflects our choices of orientations for the edges. Thus $P_k(K_n)$ is naturally an S_n-representation.

To identify this representation, let us recall the polynomial realization of the Specht modules. Let $\lambda \vdash n$, and let T be a Young tableau of shape λ, filled bijectively with $1, 2, \dots, n$. For a column of T whose entries, from top to bottom, are $i_1, \dots, i_r$, form the Vandermonde factor

$$\prod_{1 \le a < b \le r} (x_{i_a} - x_{i_b}).$$

The *Specht polynomial* of T is the product of these factors over all columns of T:

$$\Delta_T := \prod_{\substack{\text{column } C \text{ of } T; \\ i_1, \dots, i_r \text{ are the entries of } C}} \; \prod_{1 \le a < b \le r} (x_{i_a} - x_{i_b}).$$

The span of the Specht polynomials Δ_T over all tableaux T of shape λ is the usual Specht module S^λ. See, e.g., [27, Satz 1] or [13, Theorem 5.6.1].

Now let $\lambda = (n-k, k)$, where $0 \le k \le \lfloor n/2 \rfloor$. Such a diagram has exactly k columns of height 2, while all remaining columns have height 1. Hence every tableau T of shape $(n-k, k)$ determines a k-matching $M(T)$ of K_n, namely the pairs of entries occurring in its two-box columns, and we have

$$\Delta_T = \pm p_{M(T)}.$$

Conversely, every k-matching of K_n arises in this way from some tableau of shape $(n-k, k)$. Thus the span $P_k(K_n)$ of the p_M's is precisely the Specht module of shape $(n-k, k)$.

Theorem 4.1. *For every k satisfying $0 \le k \le \lfloor n/2 \rfloor$, we have, as S_n-modules,*

$$P_k(K_n) \cong S^{(n-k,k)}.$$

Consequently,

$$\mathrm{Hilb}(\mathcal{D}(K_n), t) = \sum_{0 \le k \le n/2} \left(\binom{n}{k} - \binom{n}{k-1} \right) t^k.$$

Proof. The S_n-module identification was established above. For the Hilbert series, we need to compute $\dim S^{(n-k,k)}$. Recall that the dimension of any Specht module S^λ equals the number f^λ of all standard Young tableaux of shape λ, and can be computed by the hook-length formula. Thus,

$$\dim S^{(n-k,k)} = f^{(n-k,k)} = \binom{n}{k} - \binom{n}{k-1} = \frac{n-2k+1}{n-k+1} \binom{n}{k}.$$

Together with (4), this yields the displayed Hilbert series. □

Remark 4.2 (Positive characteristic). The identification $P_k(K_n) \cong S^{(n-k,k)}$ and the hook length formula for the dimension of the Specht module are valid over any field. However, if $\mathbf{k}$ has positive characteristic, then the Specht module can fail to be irreducible. For general graphs, in contrast, even the Hilbert series can depend on the characteristic; see Remark 3.6.

The coefficients of $\mathrm{Hilb}(\mathcal{D}(K_n), t)$ form the Catalan triangle and count standard Young tableaux of the fixed shape $(n-k, k)$.

Remark 4.3 (An exterior analogue). Assume in this remark that $\mathbf{k}$ has characteristic zero. Bergeron, Chan, Soltani, and Zabrocki [3] study a strikingly parallel construction in the exterior algebra

$$R_n = \bigwedge(\xi_1, \ldots, \xi_n).$$

For a matching[1] $M = \{(i_1, j_1), \ldots, (i_k, j_k)\}$ of K_n, ordered so that $j_1 < \cdots < j_k$ and $i_r < j_r$ for every r, they set

$$\Delta_M := (\xi_{j_1} - \xi_{i_1}) \cdots (\xi_{j_k} - \xi_{i_k}) \in R_n.$$

They define a specific noncrossing matching $C(\alpha)$ for each ballot sequence α. Their Corollary 4.5 shows that the elements $\Delta_{C(\alpha)}$ indexed by all ballot sequences α (of length n with k entries equal to 1) form a basis of the degree-k exterior quasisymmetric harmonic space $EQH_{n,k}$. Thus,

$$\dim EQH_{n,k} = f^{(n-k,k)} = \dim P_k(K_n).$$

There is in fact a very explicit vector-space isomorphism between these two spaces. The same ballot sequences give noncrossing matchings $C(\alpha)$, and the commutative polynomials $p_{C(\alpha)}$ are linearly independent: with respect to the lexicographic order induced by $x_1 > \cdots > x_n$, the lexicographically smallest monomial of $p_{C(\alpha)}$ is obtained by choosing the right endpoint of every pair (i.e., the positions of all closing parentheses), and these monomials are distinct for different α. Since there are $f^{(n-k,k)}$ such ballot sequences, these polynomials form a basis of $P_k(K_n)$. Hence there is a unique isomorphism

$$\Psi_k : P_k(K_n) \longrightarrow EQH_{n,k}, \qquad p_{C(\alpha)} \longmapsto \Delta_{C(\alpha)}. \tag{11}$$

So, at the level of graded vector spaces with their preferred noncrossing bases, the commutative and exterior pictures are very close.

However, the parallels end here. The vector space $EQH_{n,k}$ is the span of the Δ_M corresponding to all noncrossing matchings M (by [3, Proposition 2.9 and Corollary 4.5]), but usually not the span of Δ_M for **all** the matchings M. In particular, it is not an S_n-representation, which should not come as a surprise because the S_n-action does not preserve noncrossingness (unless $k \leq 1$). Thus, there is no hope for an isomorphism of S_n-representations. [2]

[1] They refer to matchings as "pairings".

[2] Another false trail: There is an obvious linear isomorphism from the squarefree degree-k part of $\mathbf{k}[x_1, \ldots, x_n]$ to $\bigwedge^k \mathbf{k}^n$, sending $x_{a_1} \cdots x_{a_k}$ (with $a_1 < \cdots < a_k$) to $\xi_{a_1} \cdots \xi_{a_k}$. But this map does not send p_M to Δ_M: when the chosen endpoints are reordered, anticommutativity contributes signs which depend on the chosen term of the expansion. Thus, for example, $(x_1 - x_4)(x_2 - x_3) = x_1x_2 - x_1x_3 - x_2x_4 + x_3x_4$ is sent to $\xi_1\xi_2 - \xi_1\xi_3 - \xi_2\xi_4 + \xi_3\xi_4 \neq (\xi_1 - \xi_4)(\xi_2 - \xi_3)$.

To drive this point home, let us compare the span of the Δ_M for all k-matchings M of K_n with $EQH_{n,k}$. Let

$$W = \mathrm{span}\{\xi_i - \xi_j : 1 \le i, j \le n\},$$

the standard $(n-1)$-dimensional representation of S_n. For a k-matching M, the element Δ_M belongs to the subspace $\bigwedge^k W \subseteq \bigwedge^k \mathbf{k}^n$ (since the k factors $\xi_{j_r} - \xi_{i_r}$ of Δ_M all lie in W), and is nonzero (since the k factors of Δ_M are linearly independent). The S_n-orbit of any one such Δ_M consists, up to sign, of all Δ_Ns corresponding to all k-matchings N. Since the S_n-representation

$$\bigwedge^k W \cong S^{(n-k,1^k)}$$

is irreducible in characteristic zero, it follows that

$$\mathrm{span}\{\Delta_M : M \in \mathrm{Match}_k(K_n)\} = \bigwedge^k W \cong S^{(n-k,1^k)}. \tag{12}$$

Thus the complete commutative matching family $\{p_M : M \in \mathrm{Match}_k(K_n)\}$ spans the two-row module $S^{(n-k,k)}$, whereas the complete exterior matching family $\{\Delta_M : M \in \mathrm{Match}_k(K_n)\}$ spans the hook module $S^{(n-k,1^k)}$. Their dimensions are respectively

$$\binom{n}{k} - \binom{n}{k-1} \qquad \text{and} \qquad \binom{n-1}{k},$$

which agree only for $k = 0, 1$.

Already for $n = 4$ and $k = 2$, the difference between the complete commutative and exterior matching families is visible in the Garnir relation

$$p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0$$

in commuting variables. The three corresponding exterior products are linearly independent (they span $\bigwedge^2 W$, which has dimension 3).

Theorem 4.4. *Assume that the field* $\mathbf{k}$ *has characteristic* 0. *Then, there is an isomorphism of graded algebras*

$$\mathcal{D}(K_n) \cong \mathbf{k}[z_1, \ldots, z_n]/(z_1 + \cdots + z_n, z_1^2, \ldots, z_n^2), \qquad z_i \longmapsto \theta_i.$$

As an abstract commutative algebra, the quotient in Theorem 4.4 is the case $q = 1$ of the family

$$\mathbf{k}[z_1, \ldots, z_n]/(z_1^2, \ldots, z_n^2, (z_1 + \cdots + z_n)^q)$$

studied in detail by Jonsson Kling et al. [17]; among other things, they determine Gröbner bases and Hilbert series for the general family and give another proof of the strong Lefschetz property of the squarefree algebra. We nevertheless include the short Boolean-lattice argument needed for the present case.

Proof. The relations in the ideal $(z_1 + \cdots + z_n, z_1^2, \ldots, z_n^2)$ hold for the elements $\theta_1, \ldots, \theta_n$ by (1), so there is a surjective $\mathbf{k}$-algebra homomorphism

$$\Phi : \mathbf{k}[z_1, \ldots, z_n]/(z_1 + \cdots + z_n, z_1^2, \ldots, z_n^2) \to \mathcal{D}(K_n)$$

that sends each z_i to θ_i. We must show that this surjection Φ is an isomorphism. It clearly suffices to show that its domain and its codomain have equal Hilbert functions. Put

$$A := \mathbf{k}[z_1, \ldots, z_n]/(z_1^2, \ldots, z_n^2) \qquad \text{and} \qquad L := z_1 + \cdots + z_n \in A.$$

Note that the $\mathbf{k}$-vector space A has dimension 2^n and a basis consisting of the (residue classes of the) squarefree monomials

$$z_S := \prod_{i \in S} z_i \qquad \text{for all } S \subseteq [n].$$

We must compute the Hilbert function of $A/(L)$. This boils down to determining $\dim(A/(L))_k$ for all k. This is rather well-known; indeed, the maximal-rank statement below is the classical Boolean up-operator phenomenon underlying the strong Lefschetz property of this squarefree complete intersection; see also Stanley [28] and [17, §3.1]. We give the elementary argument here for the sake of completeness. We claim that multiplication by L – that is, the $\mathbf{k}$-linear map

$$U_r : A_r \longrightarrow A_{r+1}, \qquad f \longmapsto Lf$$

– has maximal rank in every degree (i.e., is injective for $r < n/2$ and surjective for $r \geq n/2$). It is enough to prove this over $\mathbb{R}$. Indeed, in the squarefree monomial bases, the matrix of U_r has only 0's and 1's, so a maximal minor which is nonzero over $\mathbb{R}$ is a nonzero integer and therefore remains nonzero over every field of characteristic zero.

The maps $U_r : A_r \to A_{r+1}$ for all r are the graded pieces of a $\mathbf{k}$-linear map $U : A \to A$, which we call the *up-operator* since it raises degrees by 1. Define the *down-operator* $D : A \to A$ as the $\mathbf{k}$-linear map given by

$$D(z_S) := \sum_{i \in S} z_{S \setminus \{i\}} \qquad \text{for each } S \subseteq [n].$$

This operator decreases degrees by 1; that is, $D(A_r) \subseteq A_{r-1}$ for each r.

Give the $\mathbb{R}$-vector space A the inner product for which the squarefree monomials z_S are orthonormal. Then D is the adjoint of U. A direct calculation gives, on A_r, the equality

$$DU - UD = (n - 2r)\ \mathrm{id}_{A_r}. \tag{13}$$

Indeed, for $|S| = r$, the mixed terms $z_{S \setminus \{i\} \cup \{j\}}$ occur once in both $DU(z_S)$ and $UD(z_S)$, while z_S occurs respectively $n - r$ and r times.

If $r < n/2$ and $0 \neq f \in A_r$, then

$$\begin{aligned}
\|Uf\|^2 &= \langle Uf, Uf \rangle = \langle DUf, f \rangle && \text{(since } D \text{ and } U \text{ are adjoint)} \\
&= \langle UDf + (n-2r)f, f \rangle && \text{(since } DU = UD + (n-2r)\ \mathrm{id}_{A_r} \text{ by (13))} \\
&= \langle UDf, f \rangle + (n-2r)\|f\|^2 \\
&= \|Df\|^2 + (n-2r)\|f\|^2 && \text{(again since } D \text{ and } U \text{ are adjoint)} \\
&> 0 && \text{(by positive definiteness and since } n - 2r > 0),
\end{aligned}$$

whence $Uf \neq 0$. Thus U_r is injective on the lower half of A (that is, for $r < n/2$). On the other hand, the adjoint of U_r with respect to our inner product is $D_{r+1} : A_{r+1} \to A_r$ (that is, the restriction of D to A_{r+1}); thus, the injectivity of U_r for $r < n/2$ entails the surjectivity of its adjoint D_{r+1} for $r < n/2$. Under the

complement identification $z_S \leftrightarrow z_{[n]\setminus S}$, this down-operator D_{r+1} becomes U_{n-r-1} (to make this more precise: we have $U_{n-r-1} = X_r \circ D_{r+1} \circ X_{r+1}^{-1}$, where X_k denotes the isomorphism $A_k \to A_{n-k}$ sending each z_S to $z_{[n]\setminus S}$). Hence the surjectivity of D_{r+1} we just proved entails that U_{n-r-1} is surjective for $r < n/2$. Substituting $n-r-1$ for r here, we conclude that U_r is surjective for all $r > n/2 - 1$, hence in particular for all $r \geq n/2$.

Thus we have shown that U_r is injective whenever $r < n/2$ and surjective whenever $r \geq n/2$. Hence, U_r always has maximal rank:

$$\operatorname{rank} U_r = \min\{\dim A_r, \dim A_{r+1}\} = \min\left\{\binom{n}{r}, \binom{n}{r+1}\right\}$$

(since $\dim A_k = \binom{n}{k}$). This proves the claim.

For $k = 0$, we plainly have $\dim(A/(L))_0 = 1$. Now let $k \geq 1$. Since $\dim A_k = \binom{n}{k}$, the claim we just proved shows that

$$\operatorname{rank} U_{k-1} = \min\left\{\binom{n}{k-1}, \binom{n}{k}\right\}.$$

But the degree-k component of $A/(L)$ is the cokernel of $U_{k-1} : A_{k-1} \to A_k$. Therefore

$$\begin{aligned}\dim(A/(L))_k &= \binom{n}{k} - \operatorname{rank} U_{k-1} = \binom{n}{k} - \min\left\{\binom{n}{k-1}, \binom{n}{k}\right\} \\ &= \max\left\{\binom{n}{k} - \binom{n}{k-1}, 0\right\}.\end{aligned}$$

This agrees with $\dim \mathcal{D}(K_n)_k$ by Theorem 4.1. The surjection Φ is therefore an isomorphism in every degree (since a surjective linear map between finite-dimensional vector spaces of the same dimension must be an isomorphism). □

Remark 4.5 (The presentation in positive characteristic). The characteristic-zero hypothesis in Theorem 4.4 is crucial. Over an arbitrary field, the same assignment $z_i \mapsto \theta_i$ gives a graded surjection

$$\Phi : \mathbf{k}[z_1, \ldots, z_n]/(z_1 + \cdots + z_n, z_1^2, \ldots, z_n^2) \longrightarrow \mathcal{D}(K_n).$$

Suppose now that $\operatorname{char} \mathbf{k} = p > 0$. In the squarefree monomial bases, the map

$$U_r : A_r \longrightarrow A_{r+1}$$

is the inclusion matrix between the r-subsets and the $(r+1)$-subsets of $[n]$. Wilson's diagonal form for inclusion matrices [29] shows, when $r < n/2$, that its nonzero diagonal entries over $\mathbb{Z}$ are

$$1, 2, \ldots, r+1 \qquad \text{(with positive multiplicities)}.$$

Hence U_r is injective over $\mathbf{k}$ if and only if $p > r+1$. By complementing subsets, the corresponding statement in the upper half is equivalent to surjectivity. Consequently all the maps U_r have maximal rank if and only if

$$p > \left\lceil \frac{n}{2} \right\rceil.$$

Since Theorem 4.1 shows that the Hilbert series of $\mathcal{D}(K_n)$ itself is independent of the characteristic, the surjection Φ above is an isomorphism exactly when

$$p > \left\lceil \frac{n}{2} \right\rceil;$$

for $p \leq \lceil n/2 \rceil$, its source has a larger Hilbert function than $\mathcal{D}(K_n)$ in at least one degree.

5. Arbitrary graphs

We begin with conventions. For a finite set W, let K_W denote the complete graph with vertex set W. Thus $K_{[n]} = K_n$. The graphs K_n and K_W are isomorphic whenever $|W| = n$; thus, they differ only in the labeling of the vertices. Thus, for the rest of this section, we **assume that the vertex set V of our graph G is $[n] = \{1, 2, \ldots, n\}$**, but we expect the reader to understand that all our results hold equally well, after relabeling, for graphs on any other n-element vertex set.

The spaces $P_k(G)$ can be computed in finite time: enumerate the k-matchings, expand their products in the monomial basis of $\mathbf{k}[x_v]_k$, and take their span (which is the column space of a matrix). In particular, $P_k(G) \subseteq P_k(K_n)$, since G is a subgraph of K_n. We may wonder when equality holds:

Definition 5.1. Let $0 \leq k \leq \lfloor n/2 \rfloor$. We say that a graph G on n vertices is *k-saturated* if

$$P_k(G) = P_k(K_n).$$

Problem 5.2. *Characterize the k-saturated graphs, either for a prescribed k or simultaneously for all $k \leq \nu(G)$.*

Note that this question can be restated as a question about the Specht generators Δ_T of the Specht module $S^{(n-k,k)}$: namely, we ask whether the subcollection selected by the k-matchings of G spans the whole Specht module. In contrast, the Specht matroid introduced by Wiltshire-Gordon, Woo, and Zajaczkowska [30] records the linear dependences among the classical Specht generators of a given Specht module S^λ.

5.1. **The elementary Garnir relations.** The elementary Garnir relations already give useful information. For any two vertices i and j, write $p_{ij} = x_i - x_j$, so that $p_{ji} = -p_{ij}$.

Lemma 5.3 (Elementary Garnir relations)**.** *Let N be a matching of G, and a, b, c, d be four vertices. Then*

$$p_N(p_{ab} + p_{bc} + p_{ca}) = 0, \tag{14}$$

$$p_N(p_{ab}p_{cd} - p_{ac}p_{bd} + p_{ad}p_{bc}) = 0. \tag{15}$$

Proof. Both identities follow by expanding $p_{ij} = x_i - x_j$. □

When the vertices a, b, c, d in Lemma 5.3 are distinct and are not incident to any edge of N, the equalities (14) and (15) are linear relations among matching generators of $P_{|N|+1}(K_n)$ and $P_{|N|+2}(K_n)$, respectively. These are precisely the three-vertex and four-vertex straightening relations for the two-row Specht polynomial model.

5.2. **k-saturation implies k-connectivity.** The first general obstruction is connectivity. We say that G is *k-connected* if G has more than k vertices and the graph $G - S$ is connected for every set $S \subset V$ with $|S| < k$. (In particular, the graph G is 2-connected if it is connected, has at least three vertices and has no cut-vertex.)

Proposition 5.4 (Connectivity obstruction)**.** *Let $1 \leq k \leq \lfloor n/2 \rfloor$. If G is k-saturated, then G is k-connected.*

Proof. Suppose that $S \subset V$ has size $s < k$ and that $G - S$ has at least two components $C_1, \ldots, C_m$. In the polynomial ring $R = \mathbf{k}[x_v : v \in V]$, define the ideal

$$J = (x_u - x_v : u, v \in C_i \text{ for some } i).$$

The quotient R/J is a polynomial ring with one variable for each C_i and one variable for each vertex of S. Hence a linear factor p_{uv} belongs to J exactly when u and v lie in the same component C_i.

Every k-matching of G has at most s edges incident with S. All its remaining edges lie inside the components C_i, so

$$p_M \in J^{k-s} \qquad \text{for each } M \in \mathrm{Match}_k(G). \tag{16}$$

Thus $P_k(G) \subseteq J^{k-s}$.

On the other hand, choose vertices $c_1 \in C_1$ and $c_2 \in C_2$. Since $n \geq 2k$ and $s < k$, we have $s \leq n - s - 2$, so there are enough further vertices outside S to pair each vertex of S with a distinct vertex outside $S \cup \{c_1, c_2\}$. Together with the edge c_1c_2, these give $s + 1$ pairwise disjoint edges of K_n whose linear factors $x_i - x_j$ do not belong to J. Complete them arbitrarily to a k-matching Q of K_n. We shall show that $p_Q \notin J^{k-s}$.

Choose one vertex $c_i \in C_i$ in each component and use the coordinates

$$z_i := x_{c_i}, \qquad y_u := x_u - x_{c_i} \quad (\text{for } u \in C_i \setminus \{c_i\}),$$

together with the variables x_s for $s \in S$. In these coordinates

$$R = \mathbf{k}[z_1, \ldots, z_m,\ x_s\ (s \in S),\ y_u], \qquad J = (y_u).$$

Introduce an auxiliary variable t and define the $\mathbf{k}$-algebra morphism $\sigma_t : R \to R[t]$ that fixes the z_i and the x_s and sends every y_u to ty_u. If $f \in J^d$, then $\sigma_t(f)$ is divisible by t^d.

For a linear factor $p_{uv} = x_u - x_v$, the polynomial $\sigma_t(p_{uv})$ is divisible by t exactly when u and v lie in the same component C_i; in that case it is divisible by t exactly once. Otherwise its constant term at $t = 0$ is a nonzero linear form in the z_i and x_s. Since the polynomial ring is a domain, the t-adic order of a product of such factors is exactly the number of factors that belong to J.

The matching Q contains the $s + 1$ chosen edges whose factors are not in J, so at most $k - s - 1$ of its factors belong to J. Hence $\sigma_t(p_Q)$ is not divisible by t^{k-s}, and therefore

$$p_Q \notin J^{k-s}.$$

Thus, $P_k(K_n) \nsubseteq J^{k-s}$ (since $p_Q \in P_k(K_n)$). Contrasting this with $P_k(G) \subseteq J^{k-s}$, we obtain $P_k(G) \neq P_k(K_n)$, which contradicts the assumption that G be k-saturated. This is the contradiction we need to complete our proof.

(The above argument using σ_t resembles the use of Rees algebras in the study of ideal powers.) □

For $k \geq 3$, k-connectivity is not sufficient.

Example 5.5. Let G be the triangular prism. It is 3-connected, but it has only four perfect matchings. Hence

$$\dim P_3(G) \leq 4 < 5 = \binom{6}{3} - \binom{6}{2} = \dim P_3(K_6),$$

so G is not 3-saturated.

5.3. **2-connectivity implies 2-saturation.** In degree 2 the connectivity obstruction is sharp. To prove this, we need some lemmas:

Lemma 5.6 (Two-row branching sequence)**.** *Let V be a finite set with $|V| \geq 4$, let $v \in V$, and put $V' := V \setminus \{v\}$. Every element of $P_2(K_V)$ is at most linear in x_v, so coefficient extraction at x_v defines a map $[x_v] : P_2(K_V) \to P_1(K_{V'})$. There is a short exact sequence*

$$0 \longrightarrow P_2(K_{V'}) \longrightarrow P_2(K_V) \xrightarrow{[x_v]} P_1(K_{V'}) \longrightarrow 0, \tag{17}$$

where the first (nonzero) map is the evident inclusion.

Proof. The first map is injective, and its image lies in the kernel of the last map. The last map is surjective: for distinct $c, d \in V'$, choose $a \in V' \setminus \{c, d\}$; then, up to sign,

$$[x_v](p_{va}p_{cd}) = p_{cd}.$$

The dimension formula in Theorem 4.1 gives

$$\dim P_2(K_V) - \dim P_2(K_{V'}) = |V| - 2 = \dim P_1(K_{V'}),$$

so that $\dim P_2(K_V) = \dim P_2(K_{V'}) + \dim P_1(K_{V'})$. All that remains is to apply the easy linear-algebraic lemma saying that if

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is a complex of finite-dimensional vector spaces such that f is injective and g is surjective and $\dim B = \dim A + \dim C$, then this complex is exact. □

Lemma 5.7 (A degree-two vertex)**.** *Every finite 2-connected graph contains a spanning subgraph that is edge-minimal subject to being 2-connected. Every such edge-minimal subgraph has a vertex of degree 2.*

Proof. The first assertion follows simply by deleting edges as long as 2-connectivity is preserved. For the second we use the standard ear characterization of 2-connected graphs; see [20, Theorem 25.4]. Let $R_1, \ldots, R_q$ be an ear decomposition of the edge-minimal graph H. If H is a cycle, every vertex has degree 2. Otherwise, the final ear R_q cannot have length 1: deleting that one edge would leave the union of the preceding ears, which is still a spanning 2-connected graph. Thus R_q has an internal vertex. Since no later ear exists, every internal vertex of R_q has degree exactly 2 in H. □

Lemma 5.8 (Suppressing a degree-two vertex)**.** *Let H be a 2-connected graph with $|V(H)| \geq 4$, and let $v \in V(H)$ have degree 2, with neighbors a, b. Put $V' := V(H) \setminus \{v\}$ and*

$$H^+ := (H - v) + ab, \qquad H^- := H^+ - ab = (H - v) - ab.$$

Thus H^- is obtained from $H - v$ by deleting ab if that edge was already present; if $ab \notin E(H - v)$, then simply $H^- = H - v$. Then H^+ is 2-connected and H^- is connected.

Proof. First consider H^+. It has at least three vertices. Let $w \in V'$. We claim that $H^+ - w$ is connected. Indeed, given two vertices of $H^+ - w$, choose a path between them in $H - w$. If this path uses v, then v is an internal vertex of the path, so both neighbors a, b of v are present in $H - w$ (since a and b are the only neighbors of v in H), and the path uses the segment $a - v - b$ (or $b - v - a$). Replacing this segment by the edge ab gives a path in $H^+ - w$ connecting the same two vertices. Thus $H^+ - w$ is connected. Hence deletion of any vertex leaves H^+ connected, so H^+ is 2-connected.

Now consider H^-. Since H is 2-connected, $H - v$ is connected. If $ab \notin E(H - v)$, there is nothing more to prove. Suppose instead that $ab \in E(H - v)$ and that deleting ab disconnects $H - v$. Then ab is a bridge of $H - v$. Let A and B be the two components of $(H - v) - ab$, with $a \in A$ and $b \in B$. If $A \neq \{a\}$, then in $H - a$ the nonempty set $A \setminus \{a\}$ has no edge to the rest of the graph: there is no such edge in $H - v - ab$, and the only edges incident with v are av and bv. This contradicts the connectedness of $H - a$. Hence $A = \{a\}$, and similarly $B = \{b\}$. Thus $V(H) = \{a, b, v\}$, contradicting $|V(H)| \geq 4$. Therefore H^- is connected.

For example, if $H = C_4$ and v is any vertex, then its neighbors a, b are not adjacent in H. Thus $H - v$ is a three-vertex path, H^+ is a triangle, and $H^- = H - v$ is that same path. □

Theorem 5.9 (Degree-two saturation)**.** *Let G be a graph on $n \geq 4$ vertices. Then*

$$P_2(G) = P_2(K_n) \quad \Longleftrightarrow \quad G \text{ is 2-connected.}$$

Proof. The forward implication is Proposition 5.4. Conversely, let G be 2-connected and choose, by Lemma 5.7, a spanning edge-minimal 2-connected subgraph $H \subseteq G$ and a degree-2 vertex v of H, with neighbors a, b. Put $V' := V \setminus \{v\}$ and

$$H^+ = (H - v) + ab, \qquad H^- = (H - v) - ab.$$

By Lemma 5.8, H^+ is 2-connected and H^- is connected. If $|V'| = 3$, then $H^+ = K_{V'}$; otherwise induction on n gives

$$P_2(H^+) = P_2(K_{V'}).$$

We want to prove that $P_2(G) = P_2(K_n)$. Since $P_2(H) \subseteq P_2(G) \subseteq P_2(K_V) = P_2(K_n)$, it is enough to prove $P_2(H) = P_2(K_V)$.

Recall the elementary linear-algebra fact that if

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

is a short exact sequence of vector spaces, and if $W \subseteq B$ is a subspace that contains A and surjects onto C, then $W = B$. Applying this to the short exact sequence (17) and the subspace $P_2(H)$ of $P_2(K_V)$, we see that it suffices to show that $P_2(H)$ contains $P_2(K_{V'})$ and surjects onto $P_1(K_{V'})$ under $[x_v]$.

We start with the first claim. Since $P_2(H^+) = P_2(K_{V'})$, it is enough to show that every matching polynomial $p_M \in P_2(H^+)$ belongs to $P_2(H)$. So consider a 2-matching M of H^+. If $ab \notin M$, then M is already a 2-matching of H, and thus $p_M \in P_2(H)$ follows. If $ab \in M$, then we let cd be the other edge of M (so $p_M = p_{ab}p_{cd}$); then, the triangle Garnir relation (14) gives

$$p_{ab}p_{cd} = (p_{av} + p_{vb})p_{cd} = p_{av}p_{cd} + p_{vb}p_{cd},$$

and both terms on the right come from 2-matchings of H, so the sum again belongs to $P_2(H)$. Hence

$$P_2(K_{V'}) = P_2(H^+) \subseteq P_2(H).$$

It remains to prove the surjection claim. For every edge cd of the connected graph H^-, at least one of a, b is not incident with cd. Pairing cd with the corresponding edge av or bv gives a 2-matching of H whose x_v-coefficient is, up to sign, p_{cd}. Thus, $p_{cd} \in [x_v]P_2(H)$. Therefore

$$[x_v]P_2(H) \supseteq P_1(H^-) = P_1(K_{V'}),$$

where the last equality follows from the connectivity of H^-. That is, $P_2(H)$ surjects onto $P_1(K_{V'})$ under $[x_v]$. The preceding linear-algebra observation now yields $P_2(H) = P_2(K_V)$, and hence $P_2(G) = P_2(K_n)$. □

5.4. **k-linkedness implies k-saturation.** There is also a simple sufficient condition in every degree.

Definition 5.10. A graph G is *k-linked* if, whenever $a_1, b_1, \ldots, a_k, b_k$ are distinct vertices, there exist pairwise vertex-disjoint paths Q_i joining a_i to b_i for $1 \leq i \leq k$. (Paths are defined with respect to the undirected graph G; they need not conform with the orientation we chose for the edges.)

Proposition 5.11 (Linkedness criterion)**.** *If G is k-linked, then G is k-saturated.*

Proof. Take an arbitrary k-matching $M = \{\{a_1, b_1\}, \ldots, \{a_k, b_k\}\}$ of K_n, and choose pairwise vertex-disjoint paths Q_i in G joining a_i to b_i. Along each path,

$$x_{a_i} - x_{b_i} = \sum_{uv \in E(Q_i)} \varepsilon_{uv}(x_u - x_v), \qquad \varepsilon_{uv} \in \{\pm 1\}.$$

Multiplying these k identities expresses p_M as a linear combination of products obtained by choosing one edge from each Q_i. Since the paths are vertex-disjoint, every such choice is a k-matching of G. Hence $p_M \in P_k(G)$. Thus $P_k(K_n) \subseteq P_k(G)$, and the result follows. □

Thus, for $1 \leq k \leq \lfloor n/2 \rfloor$, we have the useful chain

$$k\text{-linked} \implies k\text{-saturated} \implies k\text{-connected}.$$

The second implication is an equivalence for $k = 1, 2$, but Example 5.5 shows that it is not an equivalence for $k = 3$.

5.5. **Matching-complement implies k-saturation.** The Garnir relations also settle another useful family of graphs.

Theorem 5.12 (Matching-complement saturation)**.** *Let G be a graph on $n \geq 3$ vertices whose complement $\overline{G}$ is a matching. Then G is k-saturated for every $0 \leq k \leq \lfloor n/2 \rfloor$.*

Proof. Call the edges of $\overline{G}$ *forbidden*. We must show that $p_M \in P_k(G)$ for each k-matching M of K_n. We shall show this by strong induction on the number of forbidden edges in M. There is nothing to prove if M contains no forbidden edge. Suppose that $ab \in M$ is forbidden.

If $2k < n$, choose an unmatched vertex c. Since the forbidden edges form a matching, both ac and cb are edges of G. The triangle relation (14) gives

$$p_{ab} = p_{ac} + p_{cb}.$$

After multiplication by the factors belonging to $M \setminus \{ab\}$, this expresses p_M as a sum of two matching polynomials p_N and p_R where N and R each have one fewer forbidden edge than M. By induction, this shows that $p_M \in P_k(G)$.

If $2k = n$, then $n \geq 4$ and $k \geq 2$. Choose another edge $cd \in M$. Again the matching condition on $\overline{G}$ implies that all four cross edges ac, ad, bc, bd belong to G. The four-vertex Garnir relation (15) gives

$$p_{ab}p_{cd} = p_{ac}p_{bd} - p_{ad}p_{bc}.$$

Multiplying by the remaining factors again decreases the number of forbidden edges (by 1 or by 2). Induction completes the proof. □

Corollary 5.13 (Multipartite saturation)**.** *Let $r \geq 2$ and let*

$$G = K_{2,\dots,2}$$

be the complete r-partite graph with all r parts of size 2. Then

$$P_k(G) = P_k(K_{2r}) \qquad \text{for all } 0 \leq k \leq r,$$

and therefore

$$\mathrm{Hilb}(\mathcal{D}(G), t) = \sum_{k=0}^{r} \left(\binom{2r}{k} - \binom{2r}{k-1} \right) t^k.$$

Proof. The complement of $K_{2,\dots,2}$ is the perfect matching consisting of the r within-part pairs, so Theorem 5.12 applies. The Hilbert series then follows from Theorem 4.1. □

Remark 5.14. The restriction $r \geq 2$ is necessary: for $r = 1$ the graph has two vertices and no edges, so it is not saturated in degree 1.

For $r = 5$, Corollary 5.13 gives

$$\mathrm{Hilb}(\mathcal{D}(G), t) = 1 + 9t + 35t^2 + 75t^3 + 90t^4 + 42t^5.$$

5.6. **Matching-equivalent graphs.** The matching numbers $m_k(G)$ of G are the coefficients of the Hilbert series of $\mathcal{M}(G)$, but they do not determine the Hilbert series of $\mathcal{D}(G)$. Let us see when they do agree:

Proposition 5.15 (Forests)**.** *For every graph G, we have the equivalence*

$$\mathcal{D}(G) = \mathcal{M}(G) \quad \Longleftrightarrow \quad G \text{ is a forest.}$$

Consequently, if G is a forest, then

$$\operatorname{Hilb}(\mathcal{D}(G), t) = \sum_{k \geq 0} m_k(G) t^k.$$

Proof. The degree-one part $\mathcal{D}(G)_1$ is the row space of an oriented incidence matrix of G, whereas $\mathcal{M}(G)_1$ has the edge generators u_e as a basis. Hence

$$\dim \mathcal{D}(G)_1 = |V| - c(G), \qquad \dim \mathcal{M}(G)_1 = |E|.$$

If G is a forest, then $|E| = |V| - c(G)$, so the incidence rows span all of $\mathcal{M}(G)_1$. Thus every generator u_e belongs to $\mathcal{D}(G)$, and therefore $\mathcal{D}(G) = \mathcal{M}(G)$. Conversely, equality of the two algebras implies equality in degree one, hence $|E| = |V| - c(G)$, which is equivalent to every connected component of G being a tree. $\square$

For brevity, write

$$\mathfrak{m}_G(t) := \sum_{k \geq 0} m_k(G) t^k$$

for the matching generating polynomial of G. The next result shows explicitly what information beyond $\mathfrak{m}_G(t)$ can enter even when G has only one cycle. We recall that a graph is said to be *unicyclic* if it is connected and has exactly one cycle. Equivalently, a unicyclic graph is a graph obtained by adding an edge to a tree. Thus, any unicyclic graph has equally many vertices and edges.

Theorem 5.16 (Unicyclic graphs)**.** *Let G be a unicyclic graph, let C be its unique cycle, and put $F := G - V(C)$. Choose signs $\epsilon_e \in \{\pm 1\}$, for $e \in C$, such that*

$$\sum_{e \in C} \epsilon_e p_e = 0,$$

and put

$$c_C := \sum_{e \in C} \epsilon_e u_e \in \mathcal{M}(G).$$

Also set

$$\mathcal{P}(G) := \bigoplus_{k \geq 0} P_k(G),$$

regarded only as a graded vector space, and let

$$\rho_G : \mathcal{M}(G) \longrightarrow \mathcal{P}(G), \qquad u_M \longmapsto p_M$$

be the graded linear map defined on the matching basis of $\mathcal{M}(G)$. Then there is a short exact sequence of graded vector spaces

$$0 \longrightarrow \mathcal{M}(F)(-1) \xrightarrow{\cdot c_C} \mathcal{M}(G) \xrightarrow{\rho_G} \mathcal{P}(G) \longrightarrow 0. \tag{18}$$

Here $\mathcal{M}(F)(-1)_k = \mathcal{M}(F)_{k-1}$.

Consequently, with the convention $m_{-1}(F) = 0$, we have

$$\dim P_k(G) = m_k(G) - m_{k-1}(F) \qquad \text{for all } k \geq 0,$$

or equivalently,

$$\mathrm{Hilb}(\mathcal{D}(G), t) = \mathfrak{m}_G(t) - t\mathfrak{m}_F(t).$$

Proof. The first map in (18) is well defined because every edge of F is disjoint from every edge of C. Now, we shall show that it is injective.

Indeed, for any matching N of F, we have

$$c_C u_N = \sum_{e \in C} \epsilon_e u_e u_N = \sum_{e \in C} \epsilon_e u_{N \cup \{e\}}.$$

Thus, for any $z = \sum_N a_N u_N \in \mathcal{M}(F)$, we have

$$c_C z = \sum_N \sum_{e \in C} a_N \epsilon_e u_{N \cup \{e\}},$$

and all the matchings $N \cup \{e\}$ in this double sum are distinct. Thus the coefficient of $u_{N \cup \{e\}}$ in $c_C z$ is $a_N \epsilon_e$. Since $\epsilon_e \neq 0$, the equality $c_C z = 0$ therefore forces $a_N = 0$ for every N. Hence, multiplication by c_C really is injective on $\mathcal{M}(F)$.

The map ρ_G is surjective by the definition of $\mathcal{P}(G)$. Each matching N of F satisfies

$$\rho_G(c_C u_N) = p_N \sum_{e \in C} \epsilon_e p_e = 0.$$

Thus it remains only to prove exactness in the middle.

Choose an orientation of the edges and define a polynomial ring

$$S := \mathbf{k}[y_e : e \in E(G)].$$

Consider the graded $\mathbf{k}$-algebra homomorphism

$$\varphi : S \longrightarrow \mathbf{k}[x_v : v \in V(G)], \qquad y_e \longmapsto p_e \qquad (\text{that is, } y_{ij} \longmapsto x_i - x_j).$$

The edge differences $p_e = x_i - x_j$ of a connected graph G span a vector space of dimension $|V(G)| - 1$. Since G is unicyclic, $|E(G)| = |V(G)|$, so the space of linear dependences among the differences $p_e = x_i - x_j$, for $e = (i, j) \in E(G)$, is one-dimensional. The cycle relation $\sum_{e \in C} \epsilon_e p_e = 0$ is a nonzero such dependence, and consequently spans all of them. That is, the kernel of the degree-one part of φ is spanned by

$$\ell := \sum_{e \in C} \epsilon_e y_e.$$

It follows that

$$\ker \varphi = (\ell). \tag{19}$$

Indeed, extend ℓ to a basis of S_1; the images of the remaining basis vectors are linearly independent, so $S/(\ell)$ is the polynomial ring on those images.

For each $k \geq 0$, define a $\mathbf{k}$-vector subspace U_k of S by

$$U_k := \mathrm{span}_{\mathbf{k}}\{y_M : M \in \mathrm{Match}_k(G)\}, \qquad \text{where } y_M := \prod_{e \in M} y_e,$$

and put $U := \bigoplus_{k \geq 0} U_k$. The basis-preserving identification $u_M \leftrightarrow y_M$ identifies $\mathcal{M}(G)$ with U as graded vector spaces, and under this identification ρ_G is the restriction of φ to U, while the element c_C of $\mathcal{M}(G)$ corresponds to the element ℓ of U. Thus exactness in the middle of (18) is equivalent to

$$U \cap (\ell) = \ell\, U(F), \tag{20}$$

where

$$U(F) := \operatorname{span}_{\mathbf{k}}\{y_N : N \text{ is a matching of } F\}.$$

The inclusion $\ell U(F) \subseteq U \cap (\ell)$ is immediate. For the reverse inclusion, take a homogeneous element

$$f = \ell q \in U.$$

Every monomial occurring in f is a matching monomial (i.e., a monomial of the form y_M for a matching M of G), hence is squarefree. Fix $e \in C$. Since the coefficient of y_e in ℓ is nonzero and S is a domain,

$$\deg_{y_e}(\ell q) = \deg_{y_e} q + 1 \qquad (q \neq 0).$$

As $\deg_{y_e} f \leq 1$ (because every monomial occurring in f is squarefree), we thus obtain $\deg_{y_e} q = 0$. Doing this for every $e \in C$ shows that q involves no cycle variables (i.e., variables of the form y_e with $e \in C$).

Write

$$q = \sum_{\alpha} c_\alpha y^\alpha.$$

Because the y^α contain no cycle variables, the monomials $y_e y^\alpha$, as (e, α) varies with $e \in C$, are pairwise distinct. Hence there is no cancellation in

$$\ell q = \sum_{e \in C} \sum_{\alpha} \epsilon_e c_\alpha y_e y^\alpha.$$

Since this polynomial lies in U, every term with $c_\alpha \neq 0$ must be a matching monomial. It follows that $y^\alpha = y_N$ for a matching N of G. Moreover, N cannot meet any vertex of C: if an edge of N met $v \in V(C)$, then choosing a cycle edge e incident with v would make $y_e y_N$ a non-matching monomial. Thus N is a matching of $F = G - V(C)$. Therefore $q \in U(F)$, proving (20) and hence the exact sequence.

Taking degree k in (18) gives a short exact sequence

$$0 \longrightarrow \mathcal{M}(F)_{k-1} \longrightarrow \mathcal{M}(G)_k \longrightarrow P_k(G) \longrightarrow 0.$$

Therefore

$$\dim P_k(G) = m_k(G) - m_{k-1}(F).$$

Finally, summing over k and using $\dim \mathcal{D}(G)_k = \dim P_k(G)$ gives

$$\operatorname{Hilb}(\mathcal{D}(G), t) = \mathfrak{m}_G(t) - t\mathfrak{m}_F(t). \qquad \square$$

Proposition 5.17 (Matching-equivalent counterexamples)**.**

(a) *There are two graphs G and H with 4 vertices each such that*

$$m_k(G) = m_k(H) \quad \textit{for all } k \geq 0, \qquad \textit{but} \qquad \operatorname{Hilb}(\mathcal{D}(G), t) \neq \operatorname{Hilb}(\mathcal{D}(H), t).$$

(b) *There are two connected graphs G and H with 5 vertices each such that*

$$m_k(G) = m_k(H) \quad \textit{for all } k \geq 0, \qquad \textit{but} \qquad \operatorname{Hilb}(\mathcal{D}(G), t) \neq \operatorname{Hilb}(\mathcal{D}(H), t).$$

Moreover, G and H can be chosen with the same degree sequence.

(c) *There are two 2-connected graphs G and H with 6 vertices each such that*

$$m_k(G) = m_k(H) \quad \textit{for all } k \geq 0, \qquad \textit{but} \qquad \operatorname{Hilb}(\mathcal{D}(G), t) \neq \operatorname{Hilb}(\mathcal{D}(H), t).$$

Moreover, G and H can be chosen with the same degree sequence.

Proof. **(a)** Let

$$G_4 = K_{1,3}, \qquad H_4 = K_3 \sqcup K_1.$$

Both graphs have matching numbers

$$(m_0, m_1, m_2) = (1, 3, 0).$$

However, G_4 is connected while H_4 has two connected components, so Proposition 3.5 gives

$$\mathrm{Hilb}(\mathcal{D}(G_4), t) = 1 + 3t, \qquad \mathrm{Hilb}(\mathcal{D}(H_4), t) = 1 + 2t.$$

(b) Let G_5 be the triangle on $\{1, 2, 3\}$ with the path $3 - 4 - 5$ attached, and let H_5 be the cycle $1 - 2 - 3 - 4 - 1$ with the pendant edge $1 - 5$. Thus

$$E(G_5) = \{12, 23, 13, 34, 45\}, \qquad E(H_5) = \{12, 23, 34, 14, 15\}.$$

Both graphs are connected, both have degree sequence $(3, 2, 2, 2, 1)$, and both have matching generating polynomial

$$\mathfrak{m}_{G_5}(t) = \mathfrak{m}_{H_5}(t) = 1 + 5t + 4t^2.$$

For G_5, deleting the vertices of the unique triangle leaves the edge 45, whereas for H_5, deleting the vertices of the unique 4-cycle leaves only the isolated vertex 5. Hence Theorem 5.16 gives

$$\mathrm{Hilb}(\mathcal{D}(G_5), t) = (1 + 5t + 4t^2) - t(1 + t) = 1 + 4t + 3t^2,$$

while

$$\mathrm{Hilb}(\mathcal{D}(H_5), t) = (1 + 5t + 4t^2) - t = 1 + 4t + 4t^2.$$

Thus the two graphs have the same matching data and even the same degree sequence, but different Dunkl Hilbert series.

(c) Given a graph X on n vertices, let X^* be its cone, obtained by adjoining one new vertex adjacent to every vertex of X. A k-matching of X^* either avoids the new vertex, or consists of a $(k-1)$-matching of X together with an edge from the new vertex to one of the $n - 2k + 2$ vertices left unmatched. Hence

$$m_k(X^*) = m_k(X) + (n - 2k + 2)m_{k-1}(X). \tag{21}$$

In particular, coning preserves equality of matching numbers.

Let G_5, H_5 be the two graphs constructed in part (b), and add a universal vertex 6 to each. The resulting graphs G_5^*, H_5^* are 2-connected, have the same degree sequence $(5, 4, 3, 3, 3, 2)$, and by (21) have matching numbers

$$(m_0, m_1, m_2, m_3) = (1, 10, 19, 4).$$

By Theorem 5.9, both have $\dim P_1 = 5$ and $\dim P_2 = 9$.

The four perfect matchings of G_5^* give

$$p_{12}p_{34}p_{56}, \quad p_{12}p_{36}p_{45}, \quad p_{13}p_{26}p_{45}, \quad p_{16}p_{23}p_{45}.$$

The last three satisfy the four-vertex Garnir relation

$$p_{12}p_{36}p_{45} - p_{13}p_{26}p_{45} + p_{16}p_{23}p_{45} = 0,$$

while the first polynomial is independent of their span (for instance its $x_1x_3x_6$-coefficient is nonzero and the other three have none). Any two of the last three are independent, so $\dim P_3(G_5^*) = 3$.

The four perfect matchings of H_5^* give

$$p_{12}p_{34}p_{56}, \quad p_{14}p_{23}p_{56}, \quad p_{15}p_{23}p_{46}, \quad p_{15}p_{26}p_{34}.$$

They are independent: the monomials $x_1x_4x_5$, $x_1x_2x_5$, $x_1x_3x_4$, and $x_1x_2x_3$ occur, respectively, in only one of the four displayed polynomials. Hence $\dim P_3(H_5^*) = 4$. Therefore

$$\operatorname{Hilb}(\mathcal{D}(G_5^*), t) = 1 + 5t + 9t^2 + 3t^3, \qquad \operatorname{Hilb}(\mathcal{D}(H_5^*), t) = 1 + 5t + 9t^2 + 4t^3.$$

□

Remark 5.18 (Minimality). The examples in parts (a), (b), and (c) of Proposition 5.17 are minimal in their respective classes: among pairs of graphs on the same number of vertices with equal matching numbers but different Dunkl Hilbert series, the smallest possible number of vertices is 4 in general, 5 for connected graphs, and 6 for 2-connected graphs.

5.7. **Problems.**

Problem 5.19. *Describe the kernel*

$$I_G = \ker\left(\mathbf{k}[z_v : v \in V]/(\sum_{v\in C} z_v : C \text{ a component}) \longrightarrow \mathcal{D}(G)\right).$$

It contains z_v^2 for every v, but in general has additional graph-dependent relations.

Problem 5.20. *Find deletion recurrences for* $\dim P_k(G)$ *(say, for* $\mathbf{k} = \mathbb{Q}$*). Since matchings depend on vertex-disjointness, vertex deletion and deletion of the closed edge neighborhood are more natural here than ordinary matroid deletion–contraction.*

The results on saturation proved above leave a gap for $k \geq 3$:

$$k\text{-linked} \Longrightarrow k\text{-saturated} \Longrightarrow k\text{-connected}.$$

That is, we have a necessary combinatorial condition and a sufficient combinatorial condition for a graph to be k-saturated, but no condition that is both at once (except for $k \leq 2$).

Problem 5.21. *Is there such a condition, or is it too much to ask for?*

6. Log-concavity conjectures

Throughout this section, assume that $\mathbf{k}$ has characteristic zero. As Remark 3.6 shows, the Hilbert function can change in positive characteristic. For a finite simple graph $G = (V, E)$, write

$$H_G(t) := \operatorname{Hilb}(\mathcal{D}(G), t) = \sum_{k=0}^{\nu(G)} h_k(G)t^k, \qquad r(G) := |V| - c(G),$$

where $\nu(G)$ is the maximum size of a matching of G and $c(G)$ is its number of connected components, including isolated vertices. We set $h_k(G) = 0$ for $k > \nu(G)$. By Theorem 3.2, we have $h_k(G) = \dim P_k(G)$ for every $k \geq 0$. In particular, $h_0(G) = 1$, $h_1(G) = r(G)$, and $h_k(G) > 0$ for $0 \leq k \leq \nu(G)$: a matching of size k gives a nonzero polynomial $p_M \in P_k(G)$.

Conjecture 6.1 (Log-concavity)**.** *For every finite simple graph G, the Hilbert function of $\mathcal{D}(G)$ is log-concave. That is,*

$$h_k(G)^2 \geq h_{k-1}(G)h_{k+1}(G) \qquad \text{for } 1 \leq k < \nu(G).$$

Since the Hilbert function has no internal zeros, this conjecture would also imply its unimodality. A stronger version uses $r(G)$, the dimension of the degree-one part, as the order of normalization.

Conjecture 6.2 (Ultra-log-concavity)**.** *For every finite simple graph G, the sequence*

$$\left(\frac{h_k(G)}{\binom{r(G)}{k}}\right)_{k=0}^{\nu(G)}$$

is log-concave. Equivalently, writing $r = r(G)$, we have

$$k(r-k)h_k(G)^2 \geq (k+1)(r-k+1)h_{k-1}(G)h_{k+1}(G) \qquad \text{for } 1 \leq k < \nu(G).$$

The parameter $r(G)$ has an algebraic explanation. Omitting one vertex from each connected component leaves $r(G)$ Dunkl elements that still generate $\mathcal{D}(G)$, and each of these elements has square zero by (1). Thus $\mathcal{D}(G)$ is a graded quotient of

$$\mathbf{k}[z_1, \ldots, z_r]/(z_1^2, \ldots, z_r^2).$$

In particular, $h_k(G) \leq \binom{r}{k}$. In fact, passing to a monomial initial ideal and counting inclusions between its squarefree standard monomials gives

$$(k+1)h_{k+1}(G) \leq (r-k)h_k(G) \qquad \text{for } 0 \leq k < r.$$

Hence the normalized sequence in Conjecture 6.2 is already nonincreasing; the conjecture asks for its log-concavity as well.

The choice of normalization matters. By Theorem 4.1,

$$H_{K_5}(t) = 1 + 4t + 5t^2.$$

Normalizing its coefficients by $\binom{\nu(K_5)}{k} = \binom{2}{k}$ gives $(1, 2, 5)$, which is not log-concave, whereas normalization by $\binom{r(K_5)}{k} = \binom{4}{k}$ gives $(1, 1, 5/6)$. The same example also rules out real-rootedness of $H_G(t)$ in general. Exact rational rank computations verify Conjectures 6.1 and 6.2 for all simple graphs on at most seven vertices, up to isomorphism.

There is also a representation-theoretic strengthening of Conjecture 6.1. The group $\Gamma = \mathrm{Aut}(G)$ acts on $\mathbf{k}[x_v : v \in V]$ by $\sigma(x_v) = x_{\sigma(v)}$. This action preserves $P_k(G)$: it sends each matching product p_M to $\pm p_{\sigma(M)}$. All tensor products below are over $\mathbf{k}$, with the diagonal group action.

Conjecture 6.3 (Strong equivariant log-concavity)**.** *For every finite simple graph G and all integers $1 \leq i \leq j < \nu(G)$, there exists an $\mathrm{Aut}(G)$-equivariant injection*

$$P_{i-1}(G) \otimes P_{j+1}(G) \hookrightarrow P_i(G) \otimes P_j(G).$$

Taking dimensions and setting $i = j$ recovers Conjecture 6.1. Li's theorem [19] proves the analogous statement for the permutation representations on matchings. It does not immediately imply the conjecture above: the matching products have orientation signs under graph automorphisms and can satisfy linear relations,

and an injection before passing to a quotient need not descend to an injection afterwards.

For $G = K_n$, Theorem 4.1 identifies the conjectured injection with an S_n-equivariant injection

$$S^{(n-i+1,i-1)} \otimes S^{(n-j-1,j+1)} \hookrightarrow S^{(n-i,i)} \otimes S^{(n-j,j)} \qquad \left(1 \leq i \leq j < \left\lfloor \frac{n}{2} \right\rfloor\right).$$

Here a zero second part is omitted. Equivalently, the symmetric function

$$s_{(n-i,i)} * s_{(n-j,j)} - s_{(n-i+1,i-1)} * s_{(n-j-1,j+1)}$$

should be Schur-positive, with $*$ denoting the Kronecker (internal) product. For fixed i, j, this specialization holds for all sufficiently large n, by Gui's inequalities for reduced Kronecker coefficients indexed by one-row partitions [14, Theorem 5.4]. The conjecture asks for every admissible n.

Appendix A. Cut quotients and zonotopal algebras: a comparison

In this section, we leave the matching algebra $\mathcal{M}(G)$ behind and consider some other quotients of $\mathbf{k}[u_e : e \in E]$. We shall define analogues of the Dunkl matching subalgebra in these quotients, and state their Hilbert series. Almost all the work here has been done in [24] (sometimes implicitly), so we keep our proofs short.

Assume that the field $\mathbf{k}$ has characteristic 0. We shall work in the *square-zero edge algebra*

$$\widehat{\Phi}_G := \mathbf{k}[u_e : e \in E]/(u_e^2 : e \in E),$$

which is larger than $\mathcal{M}(G)$ because it only quotients out squares of edges but leaves products of distinct incident edges intact. We define the same signed incidence sums θ_v as before, but now inside this algebra. They generate a subalgebra

$$\mathcal{C}_G^{\mathrm{ex}} := \mathbf{k}[\theta_v : v \in V] \subseteq \widehat{\Phi}_G,$$

which is known as the external graphical zonotopal algebra (the forest version of the Postnikov–Shapiro algebra); see [24, §11], as well as the zonotopal framework in [1, 15].

We shall next see what happens when we impose further monomial relations coming from edge cuts. For a nonempty proper subset $S \subset V$, let

$$\delta(S) := \{e \in E : |e \cap S| = 1\}, \qquad d_S := |\delta(S)|, \qquad \eta_S := \prod_{e \in \delta(S)} u_e.$$

Because edges internal to S cancel in the incidence sum, we have

$$\Theta_S := \sum_{v \in S} \theta_v = \sum_{e \in \delta(S)} \varepsilon_{S,e} u_e, \qquad \varepsilon_{S,e} \in \{\pm 1\}. \tag{22}$$

Since the u_e are square-zero, the only surviving terms in the d_Sth power $\Theta_S^{d_S}$ use every cut edge $e \in \delta(S)$ exactly once. Therefore

$$\Theta_S^{d_S} = d_S! \left(\prod_{e \in \delta(S)} \varepsilon_{S,e} \right) \eta_S = \pm d_S!\, \eta_S. \tag{23}$$

In characteristic zero this shows, in particular, that every cut monomial η_S already belongs to $\mathcal{C}_G^{\mathrm{ex}}$.

The relation with the usual external and central power ideals makes the next quotient transparent. Put

$$R_V := \mathbf{k}[z_v : v \in V]/\left(\sum_{v\in V} z_v\right), \qquad \ell_S := \sum_{v\in S} z_v.$$

The standard graphical power-ideal presentations ([24, Theorem 11.1 and Corollary 10.5]) are

$$\mathcal{C}_G^{\mathrm{ex}} \cong R_V/(\ell_S^{d_S+1} : \varnothing \neq S \subsetneq V), \tag{24}$$

$$\mathcal{C}_G^{\mathrm{central}} \cong R_V/(\ell_S^{d_S} : \varnothing \neq S \subsetneq V). \tag{25}$$

Here z_v maps to θ_v. These presentations are the graphical cases of the external and central zonotopal algebras; see [24, Secs. 9 and 11] and [1, 15].

Proposition A.1 (The cut quotient inside the incidence-sum algebra)**.** *For a connected graph $G = (V, E)$ in characteristic zero,*

$$\mathcal{C}_G^{\mathrm{ex}}/(\eta_S : \varnothing \neq S \subsetneq V) \cong \mathcal{C}_G^{\mathrm{central}}.$$

Consequently, if $m = |E|$ and $r = |V| - 1$, then

$$\mathrm{Hilb}\left(\mathcal{C}_G^{\mathrm{ex}}/(\eta_S), t\right) = \mathrm{Hilb}\left(\mathcal{C}_G^{\mathrm{central}}, t\right) = t^{m-r} T_G(1, t^{-1}),$$

where T_G denotes the Tutte polynomial of G (see, e.g., [12]).

Proof. Under the presentation (24), the element $\ell_S^{d_S}$ maps, by (23), to a nonzero scalar multiple of η_S. Thus quotienting $\mathcal{C}_G^{\mathrm{ex}}$ by all η_S adds exactly the relations $\ell_S^{d_S} = 0$. The resulting presentation is (25). The Hilbert-series formula is the standard central zonotopal specialization of the Tutte polynomial [1, 15]; e.g., combine [24, Theorem 3.3 or Theorem 9.1] with [12, Theorem 3.12]. □

There is a second quotient by the same cut monomials η_S, obtained by imposing them on the *whole* square-zero edge algebra rather than only on its incidence-sum subalgebra. This is larger, and it has an elementary basis. Let

$$I_{\mathrm{cut}} := (\eta_S : \varnothing \neq S \subsetneq V) \subseteq \widehat{\Phi}_G.$$

For $F \subseteq E$, write $u_F := \prod_{e\in F} u_e$.

Proposition A.2 (The ambient cut quotient)**.** *For connected G, the set*

$$\{u_F : G \setminus F \text{ is connected}\}$$

is a basis of $\widehat{\Phi}_G/I_{\mathrm{cut}}$. Hence

$$\mathrm{Hilb}(\widehat{\Phi}_G/I_{\mathrm{cut}}, t) = \sum_{\substack{H\subseteq G;\\ H \text{ connected spanning}}} t^{m-|H|}$$
$$= t^{m-r} T_G(1, 1 + t^{-1}).$$

Proof. The squarefree monomials u_F form a basis of $\widehat{\Phi}_G$, and I_{cut} is a monomial ideal. Thus u_F survives in the quotient exactly when it is not divisible by any η_S, that is, exactly when $\delta(S) \not\subseteq F$ for every nonempty proper $S \subset V$. But a deletion set F contains a whole cut $\delta(S)$ if and only if $G \setminus F$ has no edge from S to $V \setminus S$, equivalently if and only if $G \setminus F$ is disconnected. This proves the basis statement and the first Hilbert-series formula.

For the second formula, use the corank–nullity expansion

$$T_G(1,y) = \sum_{\substack{H \subseteq G; \\ H \text{ connected spanning}}} (y-1)^{|H|-r}$$

(see, e.g., [12, Proposition 3.10]) and substitute $y = 1 + t^{-1}$. □

Thus the same cut relations produce two different but closely related objects. They fit into the diagram

$$\begin{array}{ccc} \mathcal{C}_G^{\mathrm{ex}} & \subseteq & \widehat{\Phi}_G \\ \downarrow & & \downarrow \\ \mathcal{C}_G^{\mathrm{central}} & \subseteq & \widehat{\Phi}_G / I_{\mathrm{cut}} \end{array} \tag{26}$$

(whose bottom row is injective because of [24, Corollary 10.5][3]). The lower-left algebra has total dimension $T_G(1,1)$, the number of spanning trees (by Proposition A.1). The lower-right algebra has total dimension $T_G(1,2)$, the number of connected spanning subgraphs (by Proposition A.2); its grading records the number of deleted edges. More precisely, if

$$R_G(p) = \sum_{\substack{H \subseteq G; \\ H \text{ connected spanning}}} p^{|H|}(1-p)^{m-|H|}$$

is the all-terminal reliability polynomial, then

$$\mathrm{Hilb}(\widehat{\Phi}_G / I_{\mathrm{cut}}, t) = (1+t)^m R_G((1+t)^{-1}).$$

References

[1] F. Ardila and A. Postnikov, *Combinatorics and geometry of power ideals*, Trans. Amer. Math. Soc. **362** (2010), 4357–4384, https://doi.org/10.1090/S0002-9947-10-05018-X.

[2] Y. Bazlov, *Nichols–Woronowicz algebra model for Schubert calculus on Coxeter groups*, J. Algebra **297** (2006), no. 2, 372–399.

[3] N. Bergeron, K. Chan, F. Soltani, and M. Zabrocki, *Quasisymmetric harmonics of the exterior algebra*, Canad. Math. Bull. **66** (2023), no. 3, 997–1013, https://doi.org/10.4153/S0008439523000024.

[4] L. Bedratyuk, *The graph algebra I: representation-theoretic structure*, arXiv:2606.29558, 2026.

[5] A. Berget, *Products of linear forms and Tutte polynomials*, European J. Combin. **31** (2010), 1924–1935, https://doi.org/10.1016/j.ejc.2010.01.006.

[6] A. Björner, L. Lovász, S. T. Vrećica, and R. T. Živaljević, *Chessboard complexes and matching complexes*, J. London Math. Soc. (2) **49** (1994), no. 1, 25–39, https://doi.org/10.1112/jlms/49.1.25.

[7] J. Blasiak, R. I. Liu, and K. Mészáros, *Subalgebras of the Fomin–Kirillov algebra*, J. Algebraic Combin. **44** (2016), no. 3, 785–829, https://doi.org/10.1007/s10801-016-0688-4.

[8] H. Dao and R. Nair, *On the Lefschetz property for quotients by monomial ideals containing squares of variables*, Comm. Algebra **52** (2024), no. 3, 1260–1270, https://doi.org/10.1080/00927872.2023.2260012.

[3] Indeed, the algebra $\mathcal{B}_G$ in [24] is our $\mathcal{C}_G^{\mathrm{central}}$, while the algebra Φ_G in [24] is our $\widehat{\Phi}_G / I_{\mathrm{cut}}$. Thus, by identifying $\mathcal{B}_G$ with the subalgebra $\mathcal{C}_G$ of Φ_G, [24, Corollary 10.5] shows that $\mathcal{C}_G^{\mathrm{central}}$ embeds into $\Phi_G = \widehat{\Phi}_G / I_{\mathrm{cut}}$.

[9] D. Ferrarello and R. Fröberg, *The Hilbert series of the clique complex*, Graphs Combin. **21** (2005), no. 4, 401–405, https://doi.org/10.1007/s00373-005-0634-z.
[10] S. Fomin and A. N. Kirillov, *Quadratic algebras, Dunkl elements, and Schubert calculus*, in Advances in Geometry, Progr. Math. 172, Birkhäuser, 1999, 147–182, https://doi.org/10.1007/978-1-4612-1770-1_8.
[11] Y. Filmus and N. Lindzey, *Harmonic polynomials on perfect matchings*, Sém. Lothar. Combin. **86B** (2022), Art. 59, 12 pp, https://yuvalfilmus.cs.technion.ac.il/Papers/harmonic.pdf.
[12] A. Goodall, *The Tutte polynomial and related polynomials: Lecture notes 2010, 2012, 2014*, 10 April 2017, https://iuuk.mff.cuni.cz/~andrew/VKKI.pdf.
[13] D. Grinberg, *An introduction to the symmetric group algebra*, lecture notes, 12 June 2026, https://www.cip.ifi.lmu.de/~grinberg/t/24s/sga.pdf.
[14] T. Gui, *Conjectures on the reduced Kronecker coefficients*, arXiv:2210.14668v4, 2026.
[15] O. Holtz and A. Ron, *Zonotopal algebra*, Adv. Math. **227** (2011), 847–894, https://doi.org/10.1016/j.aim.2011.02.012.
[16] J. Jonsson, *Simplicial Complexes of Graphs*, Lecture Notes in Math. **1928**, Springer, 2008, https://doi.org/10.1007/978-3-540-75859-4.
[17] F. Jonsson Kling, S. Lundqvist, F. Mohammadi, M. Orth, and E. Sáenz-de-Cabezón, *Gröbner bases, resolutions, and the Lefschetz properties for powers of a general linear form in the squarefree algebra*, J. Algebra **700** (2026), 146–184, https://doi.org/10.1016/j.jalgebra.2026.04.012.
[18] A. N. Kirillov and T. Maeno, *Noncommutative algebras related with Schubert calculus on Coxeter groups*, European J. Combin. **25** (2004), no. 8, 1301–1325.
[19] S. Li, *Equivariant log-concavity of graph matchings*, Algebraic Combin. **6** (2023), no. 3, 615–622, https://doi.org/10.5802/alco.284.
[20] M. Lavrov, *Start Doing Graph Theory*, online book, version August 1, 2026, especially §25, https://vertex.degree/.
[21] R. I. Liu, *Matching polytopes and Specht modules*, Trans. Amer. Math. Soc. **364** (2012), 1089–1107, https://doi.org/10.1090/S0002-9947-2011-05516-9.
[22] J. Migliore, U. Nagel, and H. Schenck, *The weak Lefschetz property for quotients by quadratic monomials*, Math. Scand. **126** (2020), no. 1, 41–60, https://doi.org/10.7146/math.scand.a-116681.
[23] Y. Numata, *The Lefschetz property for an algebra defined by matchings*, arXiv:2302.11039, 2023.
[24] A. Postnikov and B. Shapiro, *Trees, parking functions, syzygies, and deformations of monomial ideals*, Trans. Amer. Math. Soc. **356** (2004), 3109–3142, https://doi.org/10.1090/S0002-9947-04-03547-0.
[25] A. Postnikov, B. Shapiro, M. Shapiro, *Algebras of Curvature Forms on Homogeneous Manifolds*, in: Differential topology, Infinite-dimensional Lie algebras and applications, Amer. Math. Soc. Transl. Ser 2, vol 194, (2000), 227–235, https://arxiv.org/pdf/math/9901075.
[26] B. Shapiro, M. Shapiro, *On ring generated by Chern 2-forms on $\mathbb{SL}_n/B$*, C. R. Acad. Sci. Paris Sér. I Math. vol 326, issue 1 (1998), 75–80, https://doi.org/10.1016/S0764-4442(97)82716-4.
[27] W. Specht, *Die irreduziblen Darstellungen der symmetrischen Gruppe*, Math. Z. **39** (1935), 696–711, https://eudml.org/doc/168580.
[28] R. P. Stanley, *Weyl groups, the hard Lefschetz theorem, and the Sperner property*, SIAM J. Algebraic Discrete Methods **1** (1980), no. 2, 168–184, https://doi.org/10.1137/0601021.
[29] R. M. Wilson, *A diagonal form for the incidence matrices of t-subsets vs. k-subsets*, European J. Combin. **11** (1990), no. 6, 609–615, https://doi.org/10.1016/S0195-6698(13)80046-7.
[30] J. D. Wiltshire-Gordon, A. Woo, and M. Zajaczkowska, *Specht polytopes and Specht matroids*, in *Combinatorial Algebraic Geometry*, Fields Inst. Commun. **80**, Springer, 2017, 201–228, https://doi.org/10.1007/978-1-4939-7486-3_10.